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A Mini Project Synopsis

“onespace”

For the partial fulfillment for the award of the Bachelor of Technology (Computer Science & Information Technology) Degree by Rajiv Gandhi Proudyogiki Vishwavidhyalaya, Bhopal.

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1. Abstract

OneSpace is a web-based project-centered work context management system designed for students and individual users who manage a single project through multiple digital resources. In a typical academic workflow, research links, notes, tasks, files, documents and other project information may be distributed across different applications and locations. This fragmentation can increase context switching, make information difficult to locate and require users to repeatedly reconstruct the state of their work.

OneSpace addresses this problem by treating a project or goal as the primary organizational unit and associating its relevant tasks, notes, files, external resources and recent activity with one persistent workspace. The system provides a centralized environment in which users can access and manage project-related information through a common workspace rather than maintaining each resource independently.

The proposed system will be developed as a full-stack web application using HTML5, CSS3, JavaScript and Bootstrap 5 for the frontend, Python and Django for the backend, Django REST Framework for API development, Django ORM for database interaction, and PostgreSQL for persistent storage. Core modules include user authentication, workspace management, task management, notes, resource/link management, file management, a workspace dashboard, activity/resume information and workspace-level search.

The system is intentionally focused on project-centered work-context organization rather than attempting to replace specialized applications such as browsers, code editors or cloud platforms. External resources can be maintained through links, while resources managed directly by OneSpace remain associated with their respective workspace. The proposed system therefore provides a focused approach to organizing and maintaining project-related information within a persistent web-based workspace.

2. Introduction

Digital work is increasingly distributed across specialized applications and online services. A user working on a single project may use a browser for research, a code editor for implementation, a repository for source code, documents for requirements, a notes application for ideas, cloud storage for files, and a task-management tool for deadlines. Although each tool may be effective for its individual purpose, the resources belonging to the same project can remain scattered across separate applications and locations. This fragmentation can make it difficult to maintain an overview of the project and to quickly access the information required to continue the work.

This creates a difference between the user's mental model and the organization of digital resources. Users generally think in terms of projects, assignments, goals, or tasks, whereas software often organizes information around individual applications, files, services, or resource types. As a result, users may spend additional effort locating information, switching between tools, and manually maintaining relationships between resources that are logically part of the same project. When work is interrupted or resumed after some time, users may also need to reconstruct which resources were being used and what part of the project they were working on.

A work context in OneSpace refers to a project or goal together with the relevant resources and information required to continue that work. For example, a project workspace may contain research links, notes, tasks, uploaded documents, project files, and recent activity. The purpose is not to replace every external application, but to provide a persistent, project-centered layer in which these resources remain associated with the same context. This allows users to manage and access related information from a common workspace while continuing to use specialized external applications when required.

Existing productivity systems have reduced some aspects of information fragmentation through integrated productivity suites, knowledge-management platforms, project-management systems, and browser or operating-system workspace concepts. These approaches provide useful mechanisms for managing documents, tasks, communication, browsing resources, or collaborative work. However, their organizational models generally operate within particular platforms, applications, or categories of resources. Users may therefore still need to connect different types of project information manually.

OneSpace therefore explores a focused web-based model in which the project itself becomes the primary organizational unit for related resources. The system aims to maintain a persistent relationship between tasks, notes, files, external resources, and activity information so that project-related work can be accessed and resumed from a common context. The proposed approach focuses on resources that the web application can directly manage or reference rather than attempting universal integration with arbitrary applications and services.

3. Literature Review

Paper 1

R. Lounas, I. Hamzaoui, N. Bouguelmouna, and M. Hocine

Title: “An E-Collaboration Application for Final-Year Project Management,”

Publication: International Journal of e-Collaboration, vol. 19, no. 1, 2023.

DOI: 10.4018/IJeC.315787

Short Abstract / Summary: The paper presents an e-collaboration application designed to support final-year project management in an academic environment. The system addresses activities such as communication, coordination, production, and resource sharing among project participants. The study demonstrates how a dedicated digital environment can support different activities involved in academic project work and improve collaboration during project development.

Paper 2

R. Isa, S. Othman, A. S. Ali, and N. Azizan

Title: “Prototype Development of Final Year Project Management System to Monitor Student’s Performance,”

Publication: Journal of Advanced Research in Applied Sciences and Engineering Technology, vol. 40, no. 1, pp. 164–173, 2024.

DOI: 10.37934/araset.40.1.164173

Short Abstract / Summary: The paper presents a web-based Final Year Project Management System developed to support project monitoring and supervision in an academic environment. The system provides centralized project information and dashboard-based monitoring to help students and academic staff track project progress. The study addresses limitations associated with manual project monitoring and demonstrates the usefulness of a web-based system for improving project visibility and management.

Paper 3

A. A. Wallace, K. A. Quesenberry, and M. K. Coolsen

Title: “A Basecamp for Student Group Projects: Analysis of the Use of Project Management Software in the Advertising, Public Relations, and Marketing Classroom,”

Publication: Journalism & Mass Communication Educator, vol. 79, no. 4, 2024.

DOI: 10.1177/10776958241266891

Short Abstract / Summary: The paper examines students’ perceptions of using Basecamp as a project-management and collaboration platform for academic group projects. The study surveyed 177 students across seven course sections at two universities and investigated the usefulness of project-management software for group activities. The findings indicate generally positive student perceptions and demonstrate the potential of centralized project-management and collaboration tools in educational settings.

Paper 4

S. Thongsuk, S. Phetyoying, and C. Boonon

Title: “The Development of Project Management System, Case Study: Software Engineering, Faculty of Liberal Arts and Sciences, Sisaket Rajabhat University,”

Publication: Journal of Information Science and Technology, vol. 15, no. 1, pp. 39–48, 2025.

Short Abstract / Summary: The paper presents the development of a web-based project management system for software-engineering student projects. The system focuses on organizing and tracking project activities within a university environment and was evaluated for its effectiveness and user satisfaction. The study demonstrates the applicability of web-based project-management systems for improving the organization and monitoring of student software projects.

4. Objectives

- To centralize project-related information by bringing tasks, notes, files, external resources, and activities together within a single project workspace.
- To organize and manage project work systematically by providing features for workspace management, task management, resource organization, and collaboration.
- To improve project continuity and visibility by maintaining related project information and recent activities while providing a consolidated view of project progress.

5. Problem Statement & Need

Students and individual users frequently manage a single project across multiple disconnected tools and locations. Research material may be stored in browser tabs or PDFs, tasks in a task-management tool, notes in a notes application, files in local or cloud storage, and useful links in bookmarks or messages. Although each tool may serve its intended purpose effectively, the information belonging to the same project remains distributed across different locations. This fragmentation can cause repeated context switching, scattered information, difficulty in locating relevant resources, and difficulty in reconstructing the state of a project when work is resumed.

Existing solutions address parts of this problem through integrated productivity suites, knowledge-management platforms, project-management systems, and application-level workspaces. These systems provide useful capabilities for managing documents, tasks, collaboration, communication, or specific types of resources. However, their organizational models are generally centered on particular platforms, applications, or resource categories. Users may therefore still need to manually organize different resource types around the same project and remember where each part of their work is located.

The need addressed by OneSpace is therefore narrower than building a universal productivity platform. OneSpace aims to provide a persistent, project-centered web workspace where related tasks, notes, files, external resources, and activity information remain connected to the same project context. A user can access these resources from a consolidated workspace dashboard instead of treating each resource as an isolated item. External services are represented through links where direct integration is not required, while resources managed directly by OneSpace remain associated with their respective workspace.

The proposed solution demonstrates the project-centered work-context concept without attempting to control arbitrary desktop applications or replace specialized software. OneSpace will organize and manage resources that the web application can directly store or reference, providing a focused and practical solution for maintaining continuity and organization within a project.

6. Solution Domain

OneSpace belongs to the domain of Web Application Development, Productivity and Work-Context Management. It follows a client-server architecture in which a browser-based frontend communicates with a Python Django backend through REST-style APIs developed using Django REST Framework. The frontend provides authentication screens, workspace navigation, dashboards and resource-management interfaces using HTML5, CSS3, JavaScript and Bootstrap 5. The backend implements business logic, validation, authentication, resource management and API functionality. PostgreSQL, accessed through the Django ORM, persists the relationships among users, workspaces and their associated resources.

The central domain concept is the workspace. A workspace represents a project or goal and acts as the parent context for tasks, notes, external resources, uploaded files and activity records. This makes the workspace the primary organizational unit rather than treating each resource as an isolated item. Each resource is associated with its respective workspace so that related information can be accessed according to the project context. The workspace therefore provides a structured view of the project and its current state, while allowing different resource types to be managed through dedicated modules.

The system is designed to support common project-related activities through a unified web interface. Users can create and manage workspaces, maintain tasks and notes, store useful external links, upload project-related files and view relevant activity through the workspace dashboard. Workspace-level search further helps users locate supported resources without having to navigate through unrelated projects or resource collections. Recent activity and resume information are intended to provide basic continuity when users return to previously accessed workspaces.

The system is intentionally not designed to replace browsers, code editors, email clients, cloud-storage services or other specialized applications. Instead, it provides a persistent project-centered layer for organizing resources that the web application can directly store or reference. External resources can therefore remain in their original services while their relevant links and associations are maintained within OneSpace. This approach keeps the system focused on organizing the user's project context without requiring universal integration with external applications.

From a system-design perspective, the application separates the presentation, application and data responsibilities to support maintainability and clear development boundaries. The frontend is responsible for user interaction and presentation, Django handles application logic and request processing, Django REST Framework provides API-based communication, and PostgreSQL maintains persistent project data. This structure allows the individual modules of OneSpace to be developed and tested independently while maintaining their relationships through the central workspace model.

7. Requirement Specification (Hardware and Software Requirement) (1-2 Pages)

7.1 Hardware Requirements

- Development computer with a modern dual-core or better processor.
- Minimum **4 GB RAM; 8 GB or more recommended** for development.
- At least **10 GB available storage** for source code, dependencies, database, uploaded files and test data.
- Stable **internet connection** for development dependencies, Git/GitHub access, deployment and external resources.

7.2 Software Requirements

- **Operating System:** Windows, Linux or macOS.
- **Frontend:** HTML5, CSS3, JavaScript (ES6+) and Bootstrap 5.
- **Backend:** Python 3.x with Django.
- **API:** Django REST Framework.
- **Database:** PostgreSQL.
- **Database Layer:** Django ORM.
- **Authentication & Security:** Django Authentication and built-in Django security features.
- **Development Tools:** Visual Studio Code or equivalent IDE and a modern web browser.
- **Version Control:** Git and GitHub.
- **Deployment:** Render.

7.3 Functional Requirements

- User registration, login and logout.
- Create, view, edit, archive and delete workspaces.
- Create, update and delete workspace tasks with status, priority and due-date information.
- Create, edit, view and delete workspace notes.
- Add, edit and delete external resources or links.
- Upload, view metadata for and delete project files.
- Provide a workspace dashboard summarizing resources and progress.
- Maintain basic recent activity and resume information.
- Provide workspace-level search across supported resource types.

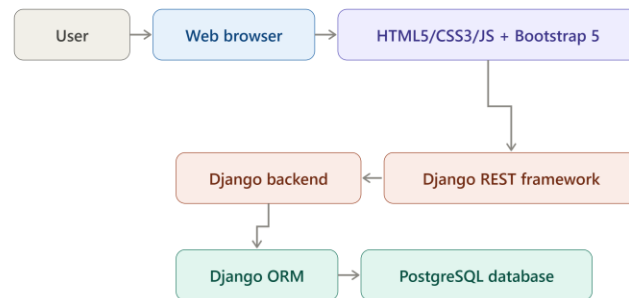
7.4 Non-Functional Requirements

- **Usability:** The interface should be simple, understandable and easy to navigate for student users.
- **Security:** Passwords must be securely handled using Django's authentication and password-hashing mechanisms, and users must only access their authorized workspaces and resources.
- **Maintainability:** Frontend, backend, API and database components should remain modular and organized.
- **Reliability:** Valid workspace data should persist correctly between user sessions.
- **Performance:** Common CRUD, dashboard and search operations should respond promptly on a basic development or hosted environment.
- **Scalability:** The database and application structure should allow additional resource types or features to be added later without major architectural changes

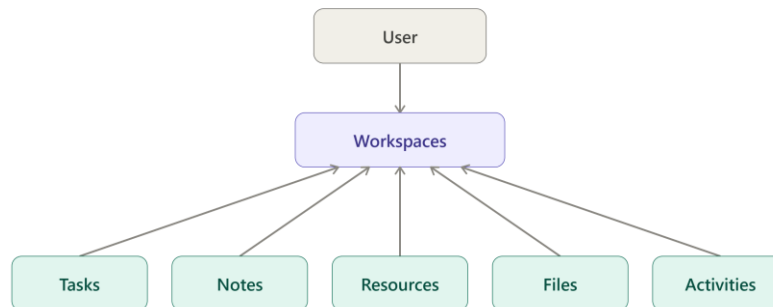
8. System Architecture Diagram

The proposed system follows a **three-layer client-server architecture**. The presentation layer uses **HTML5, CSS3, JavaScript and Bootstrap 5** to provide the user interface. The application layer uses **Django** for authentication, business logic, validation and application functionality, with **Django REST Framework** providing REST-style APIs. The data layer uses **Django ORM** with **PostgreSQL** for persistent data storage.

System Flow:



The main data relationship is:



Uploaded files will be handled by the Django application, while their **metadata, file reference and association with the respective workspace** will be maintained in the PostgreSQL database. This structure ensures that project-related resources remain associated with their respective workspace and user.

9. Tech Stack & Roadmap

9.1 Tech Stack

- **Frontend:** HTML5, CSS3, JavaScript (ES6+) and Bootstrap 5.
- **Backend:** Python with Django.
- **API:** Django REST Framework.
- **Database:** PostgreSQL.
- **Database Access:** Django ORM.
- **Authentication & Security:** Django Authentication and built-in Django security features.
- **API Communication:** JavaScript Fetch API and JSON.
- **Version Control:** Git and GitHub.
- **Deployment:** Render.
- **Development Environment:** Visual Studio Code and a modern web browser.

9.2 Development Roadmap

- **Phase 1:** Project setup, repository creation and basic Django/frontend structure.
- **Phase 2:** Database schema, Django models and user authentication.
- **Phase 3:** Workspace creation, editing, viewing, archiving and deletion.
- **Phase 4:** Task, notes, resource/link and file-management modules.
- **Phase 5:** Workspace dashboard and project progress representation.
- **Phase 6:** Activity/resume information and workspace-level search.
- **Phase 7:** Frontend refinement using Bootstrap 5, form validation, error handling and responsive design.
- **Phase 8:** API integration, functional testing, security checks, deployment, documentation, PPT preparation and final demonstration.

10. Expected Outcome

The expected outcome is a functional web application that allows users to create persistent project workspaces and organize multiple resource types within each workspace. Users will be able to authenticate, create and manage project workspaces, manage tasks, maintain notes, save useful external links, upload relevant files, and view a consolidated workspace dashboard. The system will also maintain basic recent activity and resume information so that users can identify the workspace and resource area most recently used and continue their work with less effort.

The project is expected to demonstrate how a project-centered organization model can provide a clearer and more continuous view of related digital work than managing each resource independently. By associating tasks, notes, files, external resources, and activity information with the same workspace, OneSpace will provide a structured way to access project-related information from one place. The final prototype will not attempt universal application integration or replace specialized software. External services can be represented through links, while resources supported directly by the system will be managed within the workspace.

The system will be evaluated through functional correctness, ease of navigation, persistence of workspace data, search effectiveness, resource organization, and the ability to locate and resume project-related work from a common workspace. The project will also demonstrate practical skills in frontend development, backend development, database design, REST API development, authentication, CRUD operations, file handling, search, data modeling, and full-stack web application architecture using the finalized technology stack. The implementation will follow a modular structure so that individual components can be developed, tested, and maintained independently.

The final system will provide a clear demonstration of the project-centered work-context concept while maintaining a focused implementation scope. The completed prototype can also provide a foundation for future enhancements such as richer collaboration features, notifications, external-service integrations, advanced search, and additional resource types after successful completion of the core system.

11. References

Research Papers

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